

# Vivek R

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## Technical Skills

**Programming Languages:** Python, C, SQL

**Technologies:** pandas, NumPy, Supervised Learning, Unsupervised Learning, NLP, Deep Learning, Neural Networks, Computer Vision, RAG (Retrieval-Augmented Generation)

**Frameworks:** TensorFlow, Keras, Scikit-learn, NLTK, Spacy, OpenCV, Transformers, Langchain

**Tools:** Jupyter Notebook, Google Colab, Visual Studio Code

## Experience

**Indian Institute of Technology (IIT) Madras**

**Chennai, India**

**Senior AI Engineer**

*Aug 2025 – Present*

Leading an AI-driven research project under the guidance of **Prof. Dr. N. Arunachalam**, focused on developing intelligent software for manufacturing-ready custom jewelry generation.

- Fine-tuned advanced generative models **SDXL** and **FLUX** using **LoRA (Low-Rank Adaptation)** for high-quality, style-consistent jewelry image synthesis from text prompts.
- Implemented **image semantic search** using **FashionCLIP** for automated similarity-based retrieval and design reference matching.
- Collaborated with the research team to optimize AI workflows and ensure scalability for industrial jewelry design pipelines.

**Orisys India Consultancy Services**

**Trivandrum, India**

**AI Engineer**

*Jun 2025 – Aug 2025*

Worked on enterprise-level AI projects, focusing on Retrieval-Augmented Generation (RAG) and chatbot systems for automation and customer support.

- Developed an intelligent offline chatbot, **Gramin Mitra**, that answered user queries from a 290-page bank PDF using **Retrieval-Augmented Generation (RAG)**.
- Designed a complete document pipeline including chunking, embedding with **SentenceTransformers**, and **FAISS**-based semantic search for efficient context retrieval.
- Integrated a local LLM (**Mistral via Ollama**) to generate human-like responses strictly grounded in source documents, preventing hallucinations.
- Improved response accuracy using context filtering, prompt engineering, and fallback handling, enabling production-ready customer support automation.

**Skills:** Python, FAISS, SentenceTransformers, Ollama, Mistral, PyMuPDF, RAG, NLP

## Education

**IHUB School of Learning**

**Ernakulam**

*PG Diploma in Artificial Intelligence and Machine Learning*

*2025*

**College of Engineering Muttathara**

**Trivandrum**

*Bachelor of Technology in Computer Science and Engineering, CGPA: 80.2*

*2024*

## Projects

**NUERA AI / LLM Chatbot**

*Jun 2025*

- Built a real-time voice assistant using **Whisper** for speech-to-text, **Mistral LLM (via llama-cpp)** for context-aware replies, and **Coqui TTS** for human-like speech synthesis.
- Enabled both voice and typed input in an interactive loop using **sounddevice**, **queue**, and **threading** for low-latency audio handling.
- Integrated persistent chat history with context truncation and dynamic prompt crafting for stable, coherent multi-turn conversations.
- Achieved high responsiveness and accuracy across diverse queries using local LLM and fully offline processing.

**Skills:** Python, Whisper, Llama.cpp, Coqui TTS, Threading, LLM (LLAMA3)